Alliance University
Alliance School of Advanced Computing
Alliance One -2026 - SHASTRA
CodeSangram 3.0 – Hackathon

Program Schedule

Day 1 – Thursday, 29 October 2026

Time	Phase / Session	Activity and Deliverables
09:00 AM – 09:30 AM	Inauguration & Orientation	Welcome address, event orientation, rules, and evaluation criteria.
09:30 AM – 10:30 AM	Phase 1 – Ideastorm	Announcement of problem statements and solution ideation.
10:30 AM – 11:00 AM	Refreshment Break	Morning refreshments.
11:00 AM – 01:00 PM	Phase 1 – Ideastorm (Continued)	Analyse the problem, identify target users, develop innovative concepts, and prepare pitches.
11:00 AM – 01:00 PM	**Review 1 – Problem Understanding and Ideation**	Three-minute concept pitches, problem analysis, innovation, and preliminary jury Q&A. 20 marks
01:00 PM – 02:00 PM	Lunch Break	Staggered lunch.
02:00 PM – 05:00 PM	Phase 2 – Blueprint Preparation	Initial system architecture, technology stack selection, data flow, security, scalability, and UI/UX planning.
03:00 PM – 03:30 PM	Refreshment Break	Afternoon refreshments.
05:00 PM	Dispersal	Day 1 concludes. Teams refine their blueprints for the next morning's review.

Day 2 – Friday, 30 October 2026

Time	Phase / Session	Activity and Deliverables
08:30 AM – 09:00 AM	Reassembly	Participants report to the venue.
09:00 AM – 10:30 AM	**Review 2 – Solution Blueprint and Technical Design**	Jury evaluates system architecture, technology stack, data flow, security considerations, scalability, UI/UX design, and technical feasibility. 15 marks
10:30 AM – 11:00 AM	Refreshment Break	Morning refreshments.
11:00 AM – 01:00 PM	Phase 3 – Build Sprint I	Implement core features, application logic, and functional modules.
01:00 PM – 02:00 PM	Lunch Break	Staggered lunch.
02:00 PM – 03:00 PM	Phase 3 – Build Sprint I (Continued)	Continue backend and frontend implementation and initial testing.
03:00 PM – 03:30 PM	Refreshment Break	Afternoon refreshments.
03:30 PM – 05:00 PM	Phase 4 – Build Sprint II	Integrate modules, develop the MVP, test functionality, and improve usability.
05:00 PM – 07:30 PM	Extended Development Sprint	Performance optimization, bug fixing, and mentor-guided refinements.
07:30 PM – 08:30 PM	Dinner Break	Staggered dinner.

08:30 PM – 11:00 PM	**Overnight Development Sprint**	Continued implementation, testing, and final demonstration preparation.
11:00 PM – 11:15 PM	Refreshment Break	Late-night refreshments.
11:15 PM – 12:00 AM	Continuous Build	Complete priority features and continue development.
03:30 PM – 05:00 PM	**Review 3 – Prototype Development and Progress Review**	Jury assesses core feature development, implementation progress, module integration, testing, and MVP readiness. 15 marks

Day 3 – Saturday, 31 October 2026 (Grand Finale)

Time	Phase / Session	Activity and Deliverables
01:00 AM – 03:00 AM	Overnight Sprint	Asset polishing, defect resolution, and executable packaging.
03:00 AM – 03:30 AM	Refreshment Break	Early-morning refreshments.
03:30 AM – 08:30 AM	Build Preparation	Final testing, refinements, and SDG alignment verification.
08:30 AM – 10:30 AM	Phase 5 – Prototype Demo Readiness Review	Prototype walkthrough and demonstration-readiness checks.
10:30 AM – 11:00 AM	Refreshment Break	Morning refreshments.
11:00 AM – 01:00 PM	Phase 6 – Grand Finale	Final solution polishing and presentation preparation.
10:00 AM – 01:00 PM	**Review 4 – Final Prototype Evaluation**	Jury evaluates the working prototype, live demonstration, architecture, functionality, testing, innovation, social impact, scalability, limitations, and responses to questions. 50 marks
01:00 PM – 03:00 PM	Lunch Break	Staggered lunch.
03:00 PM – 03:30 PM	Jury Deliberation	Score consolidation and final ranking.
03:30 PM – 04:00 PM	Valedictory Session	Guest addresses, feedback, and event reflections.
04:00 PM – 04:30 PM	Prize Distribution	Winner announcements, certificates, and group photographs.
04:30 PM	Dispersal	Official event closure.

Problem Statements – "Solving for Bengaluru"

Prb. No.	Problem Statement	Focus Area	Potential Ministry / Organization	SDG Alignment	Existing Solution / Current Practice	Expected Outcome of Proposed Solution
CS/26-001	AI-Based Smart Traffic Management: Develop an intelligent system to predict traffic congestion and suggest optimized routes and traffic-management strategies.	AI/ML, Smart Mobility	Bengaluru Traffic Police / Urban Transport Authorities	SDG 11 – Sustainable Cities and Communities; SDG 9 – Industry, Innovation and Infrastructure	Google Maps, VAC, Mappls, adaptive traffic signal systems	Predict congestion, recommend optimized routes, reduce travel time and support data-driven traffic management.
CS/26-002	Smart Public Transport Optimization: Develop a solution to improve public transport planning, route efficiency and passenger experience using real-time data.	Smart Mobility, Data Science	BMTC / Urban Transport Authorities	SDG 11 – Sustainable Cities and Communities	BMTC tracking, Google Maps, public transport applications	Improve route planning, reduce waiting time, optimize fleet utilization and enhance passenger experience.
CS/26-003	Intelligent Parking Management: Develop a smart parking solution to identify parking availability, reduce search time and optimize urban parking spaces.	IoT, Smart City	BBMP / Urban Local Bodies	SDG 11 – Sustainable Cities and Communities	Parking applications, sensor-based smart parking systems	Provide real-time parking availability, reduce search time and improve utilization of parking spaces.
CS/26-004	AI-Powered Waste Management: Develop a system for waste classification, collection optimization and real-time monitoring of waste-management activities.	AI/ML, Sustainability	BBMP / Urban Local Bodies	SDG 11 – Sustainable Cities and Communities; SDG 12 – Responsible Consumption and Production	BBMP waste collection systems, smart-bin solutions	Improve waste segregation, optimize collection routes, reduce overflow and support efficient waste management.
CS/26-005	Urban Flood Prediction & Management: Develop an early-warning system to predict waterlogging and urban flooding	AI, Environment	BBMP / Disaster Management Authorities	SDG 11 – Sustainable Cities and Communities; SDG 13 – Climate Action	Weather alerts, flood-monitoring and drainage systems	Provide early flood/waterlogging alerts, identify vulnerable locations and support faster disaster response.

	using weather, drainage and geographical data.					
CS/26-006	Water Conservation for Bengaluru: Develop a technology-driven solution for monitoring water consumption, detecting wastage and promoting efficient water management.	IoT, Sustainability	BWSSB / Ministry of Jal Shakti	SDG 6 – Clean Water and Sanitation; SDG 11 – Sustainable Cities and Communities	BWSSB monitoring, smart water meters, leak-detection systems	Detect wastage and leakages, monitor consumption and promote efficient and sustainable water usage.
CS/26-007	Air Quality & Pollution Monitoring: Develop an intelligent platform to monitor pollution levels and identify pollution hotspots across Bengaluru.	IoT, Green Technology	Karnataka State Pollution Control Board / MoEFCC	SDG 3 – Good Health and Well-Being; SDG 11 – Sustainable Cities and Communities; SDG 13 – Climate Action	AQI monitoring stations, CPCB/SPCB monitoring platforms	Provide real-time pollution information, identify hotspots and support targeted environmental interventions.
CS/26-008	Citizen Safety & Emergency Response: Develop a smart platform for faster reporting, location-based alerts and coordinated emergency response.	AI, Digital Governance	Bengaluru Police / Emergency Services	SDG 11 – Sustainable Cities and Communities; SDG 16 – Peace, Justice and Strong Institutions	Emergency helplines, police applications, location-based emergency services	Enable faster incident reporting, location-based alerts and improved emergency-response coordination.
CS/26-009	Smart Healthcare Access: Develop a technology solution to improve access to healthcare services, appointment management and health information for urban communities.	Healthcare, AI	Ministry of Health & Family Welfare / Health Authorities	SDG 3 – Good Health and Well-Being; SDG 10 – Reduced Inequalities	Telemedicine platforms, hospital appointment systems, ABDM services	Improve healthcare access, simplify appointment management and connect citizens with appropriate healthcare resources.
CS/26-0010	Mental Wellness Support: Develop an AI-enabled platform for early identification of mental-health concerns and connecting citizens with appropriate support resources.	AI, Healthcare	Ministry of Health & Family Welfare	SDG 3 – Good Health and Well-Being	Tele-MANAS, mental-health applications and online counselling platforms	Support early identification of concerns and connect citizens with appropriate professional support and resources.

CS/26-0011	Smart Education & Skill Development: Develop a platform to identify skill gaps and provide personalized learning and career-development opportunities.	AI, EdTech	Ministry of Education / Ministry of Skill Development & Entrepreneurship	SDG 4 – Quality Education; SDG 8 – Decent Work and Economic Growth	SWAYAM, Skill India, online learning and career platforms	Identify skill gaps, provide personalized learning recommendations and improve career-development opportunities.
CS/26-0012	Digital Citizen Services: Develop a unified platform to improve access to civic services, complaint registration, tracking and resolution.	Digital Governance	BBMP / MeitY / Urban Local Bodies	SDG 11 – Sustainable Cities and Communities; SDG 16 – Peace, Justice and Strong Institutions	Karnataka/Bengaluru online civic-service portals and complaint systems	Improve access to civic services, complaint tracking, transparency and issue-resolution efficiency.
CS/26-0013	Cybersecurity for Smart Bengaluru: Develop a solution to identify, monitor and mitigate cybersecurity risks in connected urban infrastructure.	Cybersecurity, Digital Trust	MeitY / CERT-In	SDG 9 – Industry, Innovation and Infrastructure; SDG 16 – Peace, Justice and Strong Institutions	CERT-In, SOC/SIEM platforms, cybersecurity monitoring tools	Detect cyber threats, strengthen monitoring and improve security of connected smart-city infrastructure.
CS/26-0014	Bengaluru Digital Twin: Develop an AR/VR or digital-twin-based platform to visualize urban infrastructure and support planning and decision-making.	AR/VR, Digital Twin	MoHUA / BBMP / Urban Planning Authorities	SDG 9 – Industry, Innovation and Infrastructure; SDG 11 – Sustainable Cities and Communities	GIS platforms, 3D city models and digital-twin initiatives	Create an interactive digital representation of urban infrastructure for visualization, simulation, planning and decision-making.
CS/26-0015	AI-Based Urban Infrastructure Monitoring: Develop a system to identify road damage, infrastructure issues and maintenance requirements using images, sensors or AI.	AI, Smart City	BBMP / Urban Infrastructure Authorities	SDG 9 – Industry, Innovation and Infrastructure; SDG 11 – Sustainable Cities and Communities	GIS-based road monitoring, image-based inspection systems	Automatically identify infrastructure defects, prioritize maintenance and improve response time.
CS/26-0016	Green Mobility Solutions: Develop an intelligent solution to encourage sustainable transportation, EV adoption, cycling or shared mobility.	Green Tech, Smart Mobility	Ministry of Road Transport & Highways / BMTC /	SDG 7 – Affordable and Clean Energy; SDG 11 – Sustainable Cities	EV charging platforms, bike-sharing and public transport applications	Encourage sustainable transportation, increase EV/cycling/shared-mobility adoption and reduce dependence on conventional transport.

			Transport Authorities	and Communities; SDG 13 – Climate Action		
CS/26-0017	Energy-Efficient Bengaluru: Develop an AI/IoT-based solution for monitoring and optimizing energy consumption in public or commercial infrastructure.	IoT, Green Technology	Ministry of Power / BESCOM	SDG 7 – Affordable and Clean Energy; SDG 11 – Sustainable Cities and Communities; SDG 13 – Climate Action	Smart meters, building energy-management and IoT monitoring systems	Monitor energy consumption, identify inefficiencies and recommend actions to reduce energy usage and costs.
CS/26-0018	AI for Citizen Well-Being: Develop a data-driven solution that identifies urban challenges affecting quality of life and recommends actionable interventions.	AI, Social Impact	Urban Local Bodies / Government Agencies	SDG 3 – Good Health and Well-Being; SDG 11 – Sustainable Cities and Communities	Citizen surveys, civic dashboards and data analytics platforms	Identify quality-of-life challenges and generate data-driven recommendations for targeted interventions.
CS/26-0019	Smart Bengaluru Data Platform: Develop a platform that integrates multiple urban datasets to generate actionable insights for city administrators and citizens.	Data Science, AI	BBMP / MeitY / Smart City Authorities	SDG 9 – Industry, Innovation and Infrastructure; SDG 11 – Sustainable Cities and Communities; SDG 16 – Peace, Justice and Strong Institutions	Open-data portals, GIS dashboards and government data platforms	Integrate urban datasets, generate actionable insights and support evidence-based decision-making.
CS/26-0020	Inclusive Bengaluru: Develop technology solutions that improve accessibility and participation for senior citizens, persons with disabilities and underserved communities.	Social Impact, Emerging Tech	Ministry of Social Justice & Empowerment / State Government	SDG 10 – Reduced Inequalities; SDG 11 – Sustainable Cities and Communities	Accessibility applications, assistive technologies and government accessibility services	Improve accessibility, participation and access to essential services for underserved communities.

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Evaluation Rubrics

S. No.	Criteria	Excellent	Good	Satisfactory	Needs Improvement	Max. Marks
1	Problem Understanding and Relevance to Bengaluru	Demonstrates a thorough understanding of the problem and proposes a solution directly relevant to Bengaluru's challenges (13–15)	Good understanding and clear relevance, with minor gaps (10–12)	Partial understanding and moderate relevance (5–9)	Poor understanding or unclear relevance (0–4)	15
2	Innovation and Originality	Highly innovative solution with a distinctive approach and original features (13–15)	Shows good innovation with identifiable unique features (10–12)	Limited innovation; resembles existing solutions (5–9)	Little or no originality (0–4)	15
3	Technical Implementation and Feasibility	Strong technical implementation, appropriate technology choices and realistic implementation feasibility (17–20)	Good implementation with minor technical limitations (13–16)	Partial implementation with significant gaps (7–12)	Poor implementation or major feasibility concerns (0–6)	20
4	Functionality and Prototype Demonstration	Functional prototype with core features demonstrated clearly and reliably (13–15)	Prototype works with minor issues or incomplete features (10–12)	Partially functional prototype with limited demonstration (5–9)	Non-functional prototype or inadequate demonstration (0–4)	15
5	Social Impact and SDG Alignment	Strong potential social impact with clear alignment to relevant SDGs (9–10)	Good potential impact and identifiable SDG alignment (7–8)	Limited impact or partial SDG alignment (4–6)	Minimal or unclear social impact (0–3)	10

6	Scalability and Sustainability	Highly scalable, sustainable and practical for broader implementation (9–10)	Good scalability with minor limitations (7–8)	Moderate scalability requiring further improvement (4–6)	Limited scalability or sustainability (0–3)	10
7	User Experience and Accessibility	Intuitive, user-friendly and inclusive design (5)	Good usability with minor improvements needed (4)	Basic usability with noticeable limitations (2–3)	Poor usability or accessibility (0–1)	5
8	Presentation, Documentation and Q&A	Clear presentation, complete documentation and confident, technically sound answers (9–10)	Good presentation and documentation with satisfactory answers (7–8)	Adequate presentation with gaps in documentation or answers (4–6)	Unclear presentation, poor documentation or inability to answer questions (0–3)	10
Total Marks						100

Rounds and Evaluation Process

Official criterion	Overall max	R1	R2	R3	R4
Problem Understanding and Relevance to Bengaluru	15	10	0	0	5
Innovation and Originality	15	7	0	0	8
Technical Implementation and Feasibility	20	0	8	7	5
Functionality and Prototype Demonstration	15	0	0	5	10
Social Impact and SDG Alignment	10	0	0	0	10
Scalability and Sustainability	10	0	3	3	4
User Experience and Accessibility	5	0	2	0	3
Presentation, Documentation and Q&A	10	3	2	0	5
ROUND TOTAL	**100**	**20**	**15**	**15**	**50**